CSCI4998 FINAL YEAR PROJECT · PROGRESS REVIEW

Plan2Brick

From a floor plan and one sentence to validated LEGO scenes

65/65 deterministic cells

0.970 mean coverage

4,706 library candidates

0 overlaps · 0 model calls

10 November 2026 · Xiao Chuanyi

Repository: lego-nova · Interactive dashboard: out/plan2brick/results.html

CURRENT STATUS

The pipeline is complete end-to-end

The remaining work is the user-facing demo and the baseline evaluation — not the core system.

9/9

candidate scenes green

R1–R3, three candidates each

89.4%

object requests placed

236 / 264; misses are wall-capacity drops

9,100

parts assembled

4,378 furniture + 1,569 floor + 3,153 wall bricks

0

overlaps · validation errors

0–2 sub-LDU ε notes per scene

Single entry: floor plan or preset + one-sentence brief → scene IR → three furnishing candidates → deterministic assembly → validate + three-view render. Every scene in this deck comes from that one command.

WHAT IS NEW

v1.0 single entry, natural-language brief, 3-candidate generation, deterministic solver, real LEGO brick floors and walls, local workbench.

WHY IT MATTERS

The model only touches image→IR; everything after the IR is deterministic and auditable.

STATUS

Single entry and brief delivered early (10/26); walls and workbench 10/30; demo and baseline are next.

BENCHMARK · 11/3–11/4

A frozen 18-plan × 4-brief benchmark

Pre-registered protocol, fixed splits, pool 24, K=6; missing cells are reported, never filled with duplicates.

18

plans

9 real (R1–R9) + 9 synthetic

64

applicable cells

72 planned; 8 applicability skips

384

candidates

64 cells × K=6; 0 build failures

0.868

mean coverage

room coverage 0.993; distance 0.568

DIVERSITY

Four layers: programme, placement, palette, topology. QD farthest-point selection; random baseline uses the same valid pool.

ABLATIONS

D0 / R / D3 / D4: D3 beats random on 63/64 cells (+0.0138 distance); D4 beats D3 on 64/64 (+0.111 distance, +0.771 topology).

FAILURE ACCOUNTING

P/G/S/O taxonomy: 0 planning errors, 0 overlaps, 0 check/build errors; 1,857 / 11,914 requested objects unplaced (pooled 0.844).

Sources: `docs/EVALUATION_PROTOCOL.md`, `bench/out/v13_main_table/`; library v1.4 = 58 items / 16 categories.

MULTI-MODEL COMPARISON · 11/5–11/6

Claude vs GPT: a perception gap, not a planning fix

Same frozen inputs; Claude/GPT via the aipro gateway. DeepSeek is text-only, so it is not a vision arm.

Vision perception (11 plans, 1–2 runs each)

Model	Runs	Room recall	Dim MAE	Object F1	Coverage
Claude	**22**	**0.953**	**0.163 m**	**0.781**	**0.928**
GPT	**19**	**0.932**	**0.156 m**	**0.749**	**0.938**

Claude is stronger on room/object recall; GPT is slightly better on dimensions/coverage and about 10% faster. Both build 100% downstream.

LLM-direct (text plan + brief → IR)

Model	Runs	Build	Brief-effect	Coverage
Claude	**412**	**412/412**	**83.0%**	**0.931**
GPT	**433**	**433/433**	**82.9%**	**0.941**

Both families fail the same change-room-use cases (B4). The remaining gap is planning/format, not model tier.

Takeaway: model choice moves perception metrics by a few points; it does not fix the planning failures that dominate brief compliance.

TOOL VS NO TOOL · CONTROLLED 4-CASE PILOT

The tool layer turns raw generation into reliable output

A = raw LDraw; B = AI scene IR + the same deterministic builder; P = Plan2Brick deterministic (no model).

ARM	BUILD	BRIEF-EFFECT	MEAN COVERAGE	OVERLAPS	MODEL CALLS
A · raw AI direct LDraw / CSV	0 valid LDraw	—	—	10–36 body overlaps	1 per case
B · AI + builder Claude + GPT, 845 runs	845/845	701/845 · 83.0%	0.936	0	845
P · Plan2Brick deterministic, 65 cells	65/65	65/65 · 100%	0.970	0	0

RAW AI

No model produced a valid LDraw model. DeepSeek's structured CSV became a valid 153-part model, but it was a flat pattern, not a room layout; Claude/GPT CSV requests hit upstream HTTP 500.

AI + BUILDER

The builder guarantees geometry, but model planning still fails ~17% of briefs, mainly change-room-use. More model capacity did not fix it.

PLAN2BRICK

Deterministic brief parser + fit-aware solver + validation gives 100% brief compliance, 0 overlaps and 0 model calls on the full matrix.

Safe claim: the tool layer, not a stronger model, is what makes the output buildable and semantically compliant at this task.

ASSETS + ENGINEERING REVIEW · 11/7–11/9

Long-tail assets and a reviewed solver

The same queue now generates candidate floor plans and library items when the benchmark matrix is exhausted.

1,345

accepted floor plans

7 topologies; 191–197 each

4,706

library candidates

16 categories; candidate-only pool

2,577

alias updates

deduplicated against the 58-item library

0.970

deterministic coverage

up from 0.944 after solver review

Claude + GPT review → fixes

Per-function reviews of solver, diversity, brief, scoring, worker and benchmark. Applied: band-search placement, edit source-file bug, transactional brief clauses, QD edge cases, build/count validation, DNS/network cooldown, filler cap.

Verification

`nova.py selftest` 10/10; `tests/test_modelbench.py` 20/20; full deterministic matrix 65/65 build, 65/65 brief, 0 overlaps; coverage 0.944 → 0.970.

Candidate assets are not merged into the frozen benchmark library; review the pool and promote selected entries in a versioned update.

HOW IT WORKS

One entry, deterministic after the IR

LLM only for image → scene IR; brief parsing, object selection, placement and validation are zero-token.

INPUT
Floor plan / preset + brief
Image or preset; one sentence of natural-language requirements.

›

M3 · IR
Scene IR
Rooms, dimensions, object classes and counts, layout topology.

›

M1 · LIBRARY
Object retrieval
58 items / 16 categories; text + shape search.

›

M4 · SOLVER
Deterministic assembly
Wall slots, per-room AABBs, global exclusion.

›

M5 · OUTPUT
.mpd / .ldr
Furniture + brick floor + 5-course walls; BOM-ready.

›

M6 · VERIFY
Validate + 3 views
Geometry checks, coverage, home/top/front renders.

LLM BOUNDARY

Image → IR is the only model call. `generate` reads the verified M2 cache; missing cache is an explicit error, never a fabricated IR.

DETERMINISTIC CORE

Brief patch → library selection → wall-slot placement → global exclusion. Same input, same scene; every step is inspectable.

MEASURED

9/9 green; 236/264 objects; 9,100 parts (4,378 furniture + 1,569 floor + 3,153 wall bricks); 0 cross-room overlaps; 0 real validation errors.

Pipeline implemented in `novalib/` + `nova.py`; the LDraw compiler underneath is the attributed ldraw-nova compatibility layer.

END-TO-END RESULTS

Three real floor plans, three candidates each

All nine candidates pass validation and have complete three-view renders.

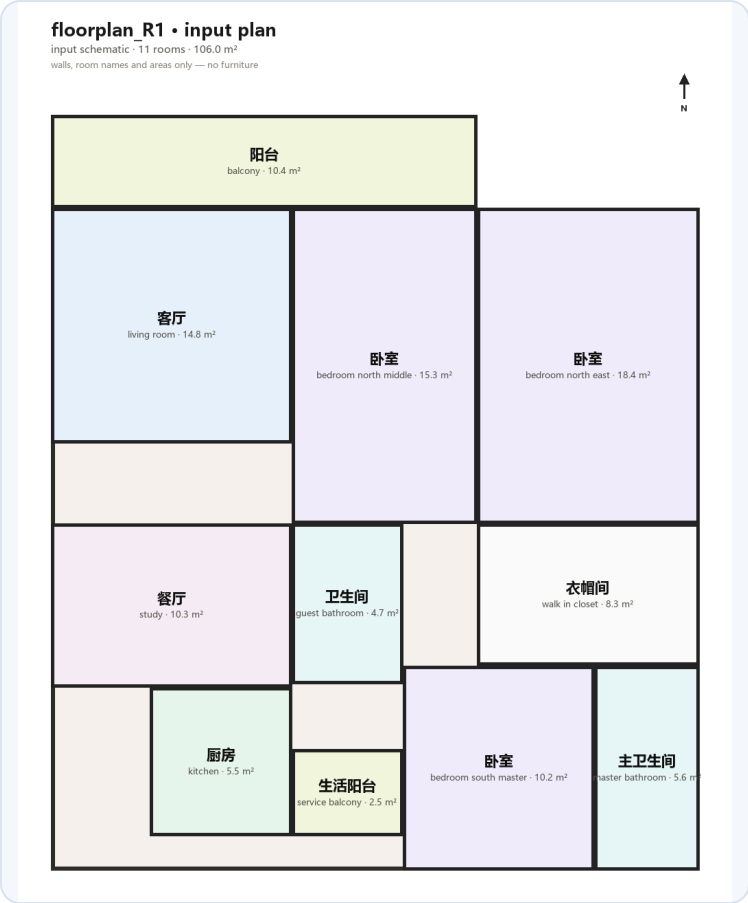
PLAN	BRIEF	COVERAGE · 3 CANDIDATES	PARTS (INCL. FLOOR + WALLS)	GATE
R1 11 rooms	Use the dining room as a study.	37/41 · 38/41 · 36/41	1,381 · 1,390 · 1,402	3/3 green ε 0/2/2
R2 6 rooms	Add a desk to the living room.	22/23 · 20/23 · 20/23	814 · 778 · 781	3/3 green ε 0/1/1
R3 7 rooms	Replace the coffee table with a desk in the living room.	21/24 · 21/24 · 21/24	855 · 844 · 855	3/3 green ε 1/1/1

Coverage misses are wall-band capacity drops, not validation failures. Parts include the 24-LDU floor slab and five courses of LEGO wall bricks; furniture-only counts are 666·675·687 / 441·405·408 / 369·358·369. ε = sub-LDU floating-point noise ignored by the gate (threshold 1e-3 LDU; real modelling errors ≥1 LDU).

Scenes green	**9 / 9**
Object coverage	**89.4%**
Parts	**9,100**
Floor bricks	**1,569**
Wall bricks	**3,153**
Cross-room overlaps	**0**
Real validation errors	**0**

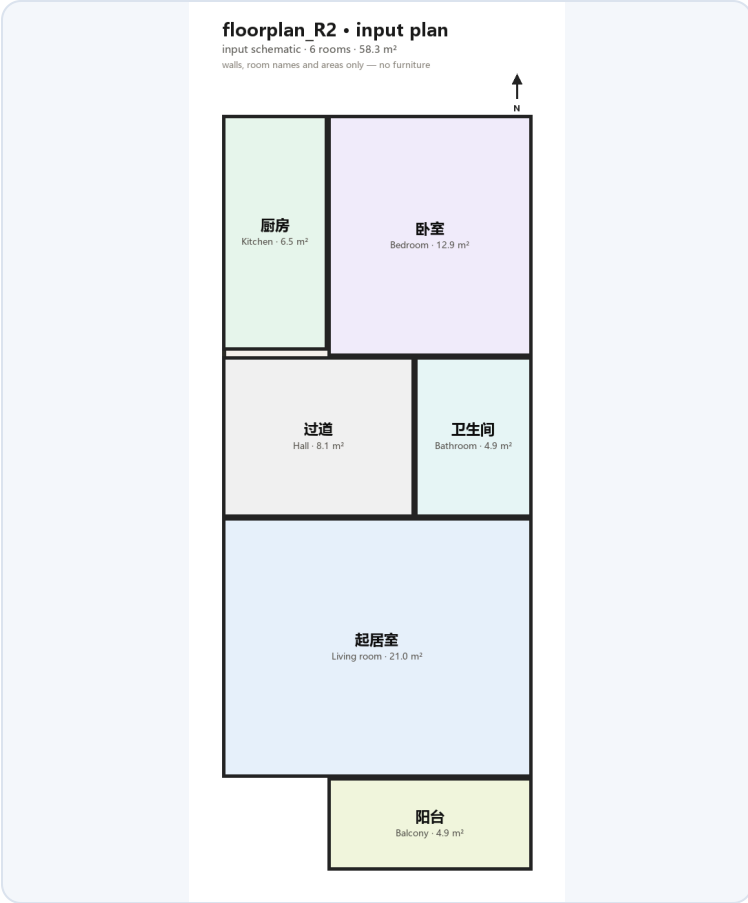
The input is a clean schematic

Walls, room names and areas only — no furniture. The same layout is then held fixed through every candidate.



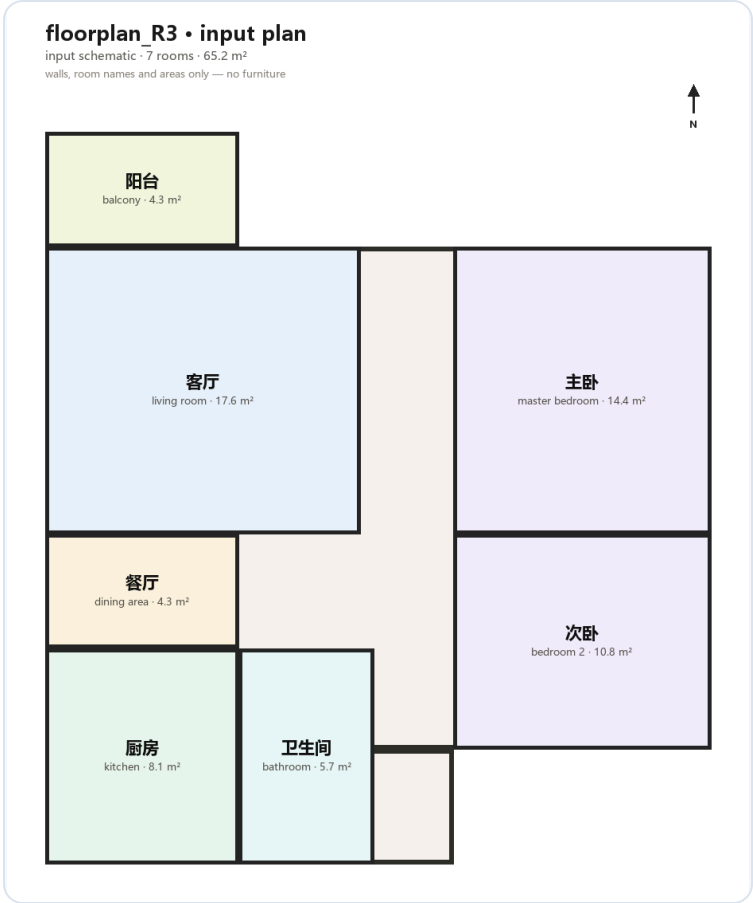
R1 · 11 rooms

106.0 m² · brief: “Use the dining room as a study.”



R2 · 6 rooms

58.3 m² · brief: “Add a desk to the living room.”



R3 · 7 rooms

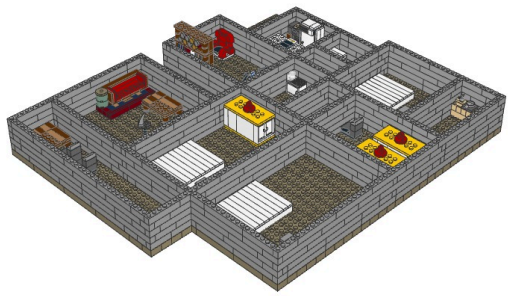
65.2 m² · brief: “Replace the coffee table with a desk.”

The clean plan is generated from the same IR the solver consumes; room outlines, bilingual names and areas come straight from the readout.

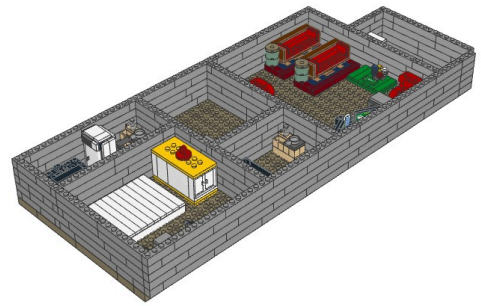
WHAT IT LOOKS LIKE · OUTPUT

The output is the same plan, built in bricks

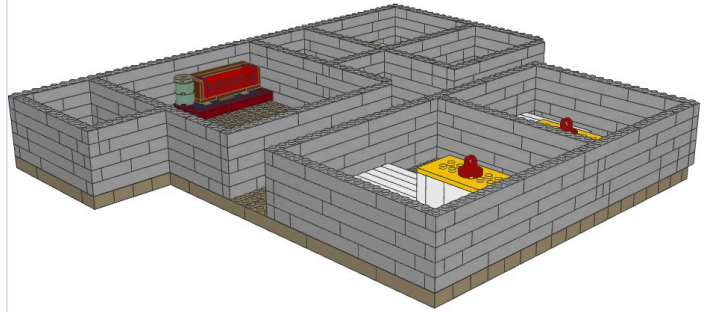
Furniture, the Dark Tan floor slab and the light-grey walls are all real LDraw bricks counted in the BOM.



R1 · candidate A
37/41 objects · 1,381 parts (666 furniture + 240 floor + 475 wall).



R2 · candidate A
22/23 objects · 814 parts (441 furniture + 124 floor + 249 wall).



R3 · candidate A
21/24 objects · 855 parts (369 furniture + 159 floor + 327 wall).

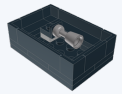
All nine candidates are green with 0 cross-room overlaps; open `results.html` to switch plans, candidates and views.

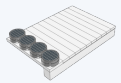
M1 · OBJECT LIBRARY

58 items across 16 categories — fully self-checked

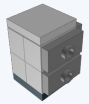
Every item validates, renders, and is retrievable by description or by shape.

42 / 16	58/58	58/58	58/58	58/58	56/56
items / categories	validate pass	three-view QC	text self-retrieval	shape self-retrieval	synonym top-1

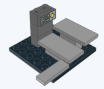

Bath


Bed


Bookshelf


Cabinet


Chair

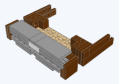

Desk

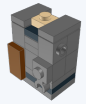

Fridge

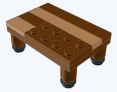

Lamp

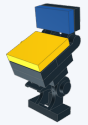

Plant


Sink


Sofa


Stove


Table


Toilet


TV


Washer

TEXT RETRIEVAL

`library --text "couch"` routes to the sofa category through captions, specs and alias phrases; ranking is auditable.

SHAPE RETRIEVAL

`library --shape 12x8x2.4 --tol .15` returns bed-9650 first (footprint error 0.0049 studs).

WHY IT MATTERS

Selection is deterministic and zero-token; no model call at this layer.

Models are sourced from BrickNet; selection, specs, captions repair, QC renders, synonyms and retrieval are the project's own. No near-duplicates; every item has five captions.

Dimensions are exact; image errors are read-off errors

Synthetic layouts isolate the solver; real layouts isolate the vision step.

Synthetic suite A / B / C

Plan A	**8 / 8 rooms**	**MAE 0.00 m**
Plan B	**6 / 6 rooms**	**MAE 0.00 m**
Plan C	**9 / 9 rooms**	**MAE 0.00 m**

Geometry solver recovers every room and dimension; 0 cross-room overlaps after assembly.

Real plans R1–R3 × image / text

Inputs	**6 / 6 green**	**0 overlaps**
Text arm	**exact dimensions**	**—**
Image arm	**read-off residuals**	**e.g. 5.1 vs 3.8 m**

R1 mid-bedroom depth is read as 5.1 m instead of 3.8 m; the solver places the room correctly from the IR it is given.

Why this matters: the residuals separate vision error from solver error. A stronger vision model can be measured against the same six-cell matrix; R3 calibration was corrected from 66.4 to 54.3 ppm before the final runs.

Synthetic suite: rooms and dimensions are generated, so the solver can be checked exactly. Real suite: image/text pair per plan; both arms produce valid scenes.

CONTROLLED STUDY · RUN 7

14 runs: no semantic gap, large cost and reliability gap

Two model tiers × with/without library + IR scaffold × image/text × three layout complexities.

LAYOUT	SCAFFOLDED TOKENS	NO-SCAFFOLD TOKENS
A	33–40k	148–159k
C	32–38k	133–209k
B	31–35k	138–261k

Model-side token ranges across flash / pro[1m]. Overall scaffold reduction: 3.5–6.5× (reported range across all units).

SEMANTICS

0.00 m

Dimension MAE in every scaffolded cell; no measurable semantic gap at these complexities.

COST

3.5–6.5×

Lower model-side tokens with the library + IR scaffold.

RELIABILITY

0

Tool-level build failures in scaffolded arms; no-scaffold arms repeatedly fail builds.

COMPLEXITY

80.6 vs 90.9%

Coverage, pro vs flash on the denser B IR — the gap grows with complexity.

Honest boundary: the Astra channel has been unavailable since 2026-10-12. The visual V/T experiment is a capability proxy, not a model-to-model Astra comparison; no equivalence is claimed.

CONTRIBUTION BOUNDARY

What is mine, and what is referenced

Built in this project

- **Floor-plan layer:** scene IR, brief parser, deterministic solver, variants, single-item edit.
- **Object library:** 58-item curation, specs, QC, synonyms, text + shape retrieval.
- **Evaluation:** coverage, cross-room overlap, dimension MAE, epsilon gate, three-view QC.
- **Experiments:** Run 7/8, gallery parity, Astra-proxy, Plan2Brick v1.0 results.
- **Engineering:** ldraw-nova compatibility reimplementation (attributed; not claimed as research).

Referenced / external

- **ldraw-nova** — Carlos Antelo, AGPL-3.0; 29 modules / 31 commands reimplemented.
- **LDraw parts library** — part geometry, CCAL terms.
- **BrickNet** — object models and captions; dataset not ours.
- **LeoCAD** — rendering and BOM cross-check.
- **DeepSeek flash / pro[1m]** — model backends; Astra channel unavailable.
- **BrickWright / BrickBench** — related work for positioning, no code reuse.

Safe claim: a floor-plan → validated-scene system with its own library, solver and evaluation. **Not claimed:** ldraw-nova as original work; Astra equivalence; live vision; physical validity; SOTA against BrickWright/BrickBench.

Full attribution: `lego-nova/ATTRIBUTION.md`; upstream license: AGPL-3.0 (`LICENSE`); data/tool terms remain with their owners.

WHAT REMAINS

Three deliverables, three dates

10/31
Mid-term pipeline
done 10/26

11/08
Local demo
2-min recording

11/18
Baseline
evaluation

11/25
Freeze results
report draft

12/02
Term-End
package

11/08 · Local demo

Upload/select plan + brief → three candidates → 3D compare → edit one object → export .ldr/.mpd + BOM. Acceptance: a complete 2-minute recording; other objects unchanged after edit.

11/18 · Baseline evaluation

A/B/C + R1–R3 + six new real plans. Random placement, LLM-direct, and ablations (no library / no IR / no solver). Coverage, collisions, dimension MAE, semantic match, cost/time.

12/02 · Term-End

Report + reproducible artifact + demo video. Numbers traceable to manifests and evaluate JSON. No new gap/gallery/proxy experiments; scope is frozen.

Early delivery: the original 10/28 milestone (single entry + natural-language brief + R1–R3 runs) was completed on 10/26, so the next block is demo and evaluation.

RISKS & LIMITATIONS

Known limits, stated up front

These are the places where a careful reviewer should push — and the answer we will give.

LIMITATION	WHAT IT MEANS / MITIGATION
Cached image → IR	`generate` reads the verified M2 cache; demo is offline-safe. Live vision is optional and fails explicitly if no cache exists — we do not fabricate an IR.
Abstract rectangular plans	No wall thickness, door openings or load-bearing structure. Stated as a limitation; consistent with the prior suite and the current IR contract.
Astra channel unavailable	Only a capability proxy exists. We report it as a proxy and make no equivalence claim.
Edit does not reflow	Replaced objects keep their centre; the demo performs one edit and validates afterwards. The other objects are line-verified unchanged.
Coverage is 89.4%	The missing 28 requests are wall-band capacity drops, not invalid scenes. Both coverage and validity are reported; no 100% claim.
Shape retrieval is geometric	L×S×H + tolerance + aspect/area; good for “find a 12×8×2.4 plate”, not free-form semantic shape matching.

Reporting rule: every scene reports coverage and validity separately; every number in this deck has a manifest or evaluate JSON behind it. We do not trade one metric for the other.

`physical_validity` remains `not_proven`: validate + geometry + BOM checks are not a physics simulation. The safe wording is “passes validation and renders”, not “guaranteed buildable”.

DISCUSSION

Three decisions I need from you

Everything else is on track; these are the choices that change the 11/08–11/18 scope.

1 **Is the demo scope enough?**

Upload/select plan + brief → three candidates → 3D compare → one edit → export .ldr/.mpd + BOM, with a 2-minute recording. Would you add or drop anything?

2 **Is the baseline set right?**

Random placement, LLM-direct, and ablations (no library / no IR / no solver). Is any baseline missing that a reviewer would expect?

3 **Is the evaluation set enough?**

A/B/C + R1–R3 + six new real plans; a 3-person pilot instead of a full user study. Should we add more plans or more people?

4 **Any Astra update?**

If the channel is still unavailable, we report the capability proxy only and keep the equivalence question for Term 2.

Next check-in: around 11/08, with the 2-minute demo recording and the first baseline table. I will send `slides.pdf` and the standalone results page after this meeting.

APPENDIX

Everything is traceable

RESULTS DASHBOARD
`lego-nova/out/plan2brick/results.html`

STANDALONE COPY
`lego-nova/out/plan2brick/results_standalone.html`

RESULT SUMMARY
`lego-nova/out/plan2brick/SUMMARY.md`

FULL RUN RECORD
`lego-nova/RESULTS.md`

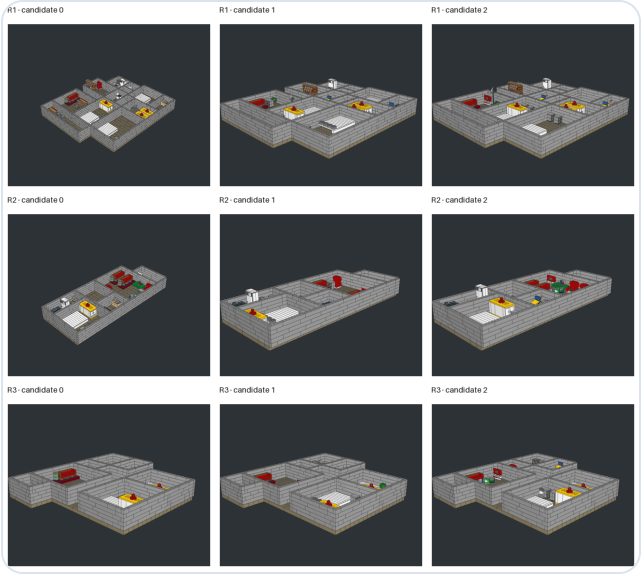
RUN 7
`lego-nova/out/gap_report.md`

LIBRARY EVALUATION
`fyp-pipeline/library_qc/EVALUATION.md`

ATTRIBUTION
`lego-nova/ATTRIBUTION.md`

SIX-WEEK PLAN
`google/own_project_plan_1022_zh.md`

Shared document: `google/main.docx` now starts with “Xiao (10/28): Current results (consolidated)”, covering the same numbers as this deck.



9/9 candidates — contact sheet

Thank you — questions?